



Lecture 3 – Code Review

30 MCQs (Dr. Ehab Essa Style)

1. Code review is mainly used to improve:

- a) Execution speed
- b) Software quality
- c) Memory usage
- d) Hardware performance

Answer: b

2. Code review is performed:

- a) After deployment
- b) Before testing
- c) Before merging code
- d) During execution

Answer: c

3. Which is NOT a goal of code review?

- a) Bug detection
- b) Knowledge sharing
- c) Style improvement
- d) Program execution

Answer: d

4. Reviewing code without running it is an example of:

- a) Dynamic analysis
- b) Code review

- c) Profiling
- d) Testing

Answer: b

5. Code review mainly focuses on:

- a) Hardware
- b) Logic and style
- c) User interface
- d) Network issues

Answer: b

6. Which of the following is a benefit of code review?

- a) Faster compilation
- b) Reduced bugs
- c) More memory
- d) Larger codebase

Answer: b

7. Code review helps new team members by:

- a) Reducing tests
- b) Sharing knowledge
- c) Removing bugs
- d) Increasing speed

Answer: b

8. Which activity is part of manual code review?

- a) Running tests
- b) Reading code
- c) Profiling
- d) Debugging

Answer: b

9. Code review should focus on:

- a) Who wrote code
- b) Code correctness
- c) Personal style
- d) Speed only

Answer: b

10. A good code review comment should be:

- a) Personal
- b) Vague
- c) Constructive
- d) Long

Answer: c

11. Which is a common code review issue?

- a) Syntax error
- b) Poor naming
- c) Runtime crash
- d) Hardware fault

Answer: b

12. Code review is MOST effective when done:

- a) At end of project
- b) Frequently
- c) Once only
- d) After deployment

Answer: b

13. Which is NOT a code review type?

- a) Pair programming
- b) Tool-based review
- c) Formal inspection
- d) Dynamic testing

Answer: d

14. Pair programming is a form of:

- a) Testing
- b) Debugging
- c) Code review
- d) Profiling

Answer: c

15. Reviewing very large changes usually:

- a) Is effective
- b) Misses issues
- c) Is faster
- d) Finds more bugs

Answer: b

16. Code review tools help mainly with:

- a) Execution
- b) Collaboration
- c) Compilation
- d) Deployment

Answer: b

17. Which is an example of code review feedback?

- a) Test fails
- b) Rename variable
- c) Program crashes
- d) Slow execution

Answer: b

18. Code review should avoid:

- a) Suggestions
- b) Questions
- c) Personal criticism
- d) Explanations

Answer: c

19. Code review can detect:

- a) Logic errors
- b) Hardware faults
- c) Network issues
- d) Power failures

Answer: a

20. One main limitation of code review is:

- a) Needs tools
- b) Time consuming
- c) Requires execution
- d) Needs compiler

Answer: b

21. Code review improves maintainability by:

- a) Reducing files
- b) Improving readability
- c) Increasing speed
- d) Adding features

Answer: b

22. Which factor affects code review quality most?

- a) Reviewer focus
- b) Compiler version
- c) Hardware
- d) IDE

Answer: a

23. Code review comments should focus on:

- a) Author
- b) Code
- c) Deadline
- d) Manager

Answer: b

24. Which is NOT checked during code review?

- a) Naming conventions
- b) Logic clarity
- c) Runtime performance
- d) Code duplication

Answer: c

25. Code review helps prevent bugs:

- a) After release
- b) Early

- c) At runtime
- d) After crash

Answer: b

26. A reviewer should mainly act as:

- a) Judge
- b) Teacher
- c) Collaborator
- d) Manager

Answer: c

27. Reviewing code line by line is called:

- a) Debugging
- b) Inspection
- c) Testing
- d) Profiling

Answer: b

28. Code review tools usually integrate with:

- a) Hardware
- b) Version control
- c) Operating system
- d) Database

Answer: b

29. Which practice improves code review effectiveness?

- a) Large commits
- b) Small changes
- c) Late reviews
- d) No comments

Answer: b

30. Code review primarily supports:

- a) Correctness
- b) Security only
- c) Performance only
- d) UI design

Answer: a



Lecture 3 – Code Review

Essay Questions (Dr. Ehab Essa Style)

Essay Question 1

Define code review. Explain its main goals in software construction.

Model Answer:

Code review is the process of **systematically examining source code** by developers **other than the author**.

It is performed **without executing the program**.

The main goals of code review are:

- Detecting **bugs and logic errors**
 - Improving **code quality and readability**
 - Ensuring **coding standards** are followed
 - Facilitating **knowledge sharing** among team members
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Essay Question 2

Explain the benefits of code review and its limitations.

Model Answer:

Benefits of code review:

- Early detection of defects
- Improved **maintainability**
- Better **code consistency**
- Reduced cost of fixing bugs
- Improved team communication

Limitations of code review:

- Time consuming
 - Depends on reviewer experience
 - May miss runtime-related bugs
 - Not a replacement for testing
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Essay Question 3

Describe best practices for effective code review.

Model Answer:

Effective code review follows these best practices:

- Review **small and focused changes**
- Provide **constructive and respectful feedback**

- Focus on **code, not the developer**
- Check **logic, readability, and style**
- Perform reviews **frequently**

These practices improve review quality and reduce missed defects.